

SUNGHO LEE

Automotive Quality Engineer · New Product Launch · Supplier APQP

5 yrs 8 mos total · 5 yrs 3 mos in automotive quality (Sep 2026, excl. internship)

Daejeon, Republic of Korea

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Korean (Native) · English (OPIc IM2)

Automotive quality engineer with ~5 years at Tier-1 suppliers (Valeo, Hanon Systems), focused on getting quality right **before SOP**: FMEA-driven launch quality for a new vehicle program, supplier APQP from process validation to line try-out, and data-level root cause analysis of field claims down to IC/EEPROM logic. Now leading supplier development in step with vehicle development gates at Hanon Systems.

CORE STRENGTHS

New Program Launch · APQP

Gate-based risk reduction from proto to SOP; FMEA, Control Plan, error-proofing

Customer & Field Quality

Led joint investigations with customer and IC maker; KRW 26M claim defended

Automotive Electronics

PCBA, Hall IC (TLE4997) EEPROM, FCT/EOLT processes, sensor products

Root Cause Analysis

FTA · PDCA · 8D; data-sheet and log-based analysis, reproduction tests

Audit & QMS

Led IATF 16949 and Valeo V5000 audits (2023, 2024); supplier process audits

Work Automation

Built my own tools with AI assistance: project dashboard, PDF auto-signing, SAP entry automation

EXPERIENCE

Hanon Systems

05/2026 – Present (5 mos)

Senior Professional, Parts & Supplier Development (P&SD), Purchasing Division · Daejeon · Full-time

- **Supplier development aligned with vehicle development phases** for heat-exchanger parts (chiller, condenser, radiator): drive quality and delivery readiness at each development gate.
- **JG program chiller development follow-up**: track development progress, check gate-by-gate quality and delivery issues, and assess supplier capacity (CAPA) and processes to confirm SOP readiness.
- **Supplier Communication Seminar (H1 2026)**: planned and hosted for ~200 supplier CEOs, delivering customer (OEM) messages such as Zero Defect and the QCD improvement direction.
- **Supplier quality KPIs**: structured supplier-level quality indicators into a management baseline and shared best-practice improvement cases across suppliers.

Work efficiency & automation (built with AI tools)

- **Supplier development dashboard**: with no dedicated system, built a dashboard for Proto–SOP schedules, gate checkpoints and issue history by vehicle and supplier; ran on Google Sheets (Apps Script) and Excel + PowerShell, then a web version on Vercel + Supabase (DB/API, RLS access control, HTTPS).
- **PDF signing automation**: Power Automate downloads PDFs, a linked Python program signs them, and the flow sends the mail.
- **SAP entry automation** with Excel VBA (QI, QDL, QP quality data, incoming B/L) · **automatic weekly request mails** to 18 suppliers.

Valeo Mobility Korea

01/2025 – 05/2026 (1 yr 4 mos)

Quality Engineer (Site SQA) · Daegu · Full-time · New legal entity and new plant after ADAS spin-off

ROA new program launch — preventive launch quality

- Rated launch risks with FMEA at the design stage: **mixed-part shipment (Middle)** and **connector cosmetic damage (High)**.
- Proposed an **error-proof labeling logic** (label prints only when barcode part number and quantity match); won approval from manufacturing engineering by quantifying the loss (~KRW 5M per mix-up, recurring at least quarterly). Added **connector vision inspection**.
- At proto stage, resolved **bracket noise** with R&D by tightening drawing tolerance (**0.3 → 0.1 mm**) and adding ribs; specified heat-staking and assembly-interference gauges to set clear acceptance criteria.
- Result: FMEA risks lowered to **Low** before SOP; stable early-production assembly and function quality.

New plant QMS & incoming quality

- Built an **SAP–Apriso integrated interlock** that blocks goods receipt until inspection results are entered; re-set key control dimensions, renewed supplier inspection agreements, and digitized ISIR/inspection records to meet Valeo 5 Star requirements.
- Cut in-process scrap rate by **13%** through incoming defect prevention.
- Solved PBT GF30 housing brittleness at an injection-molding supplier (resin residence during machine stops); purging interlock and destructive check lowered FMEA risk from High to Low, rolled out to all molding lines.

Valeo Electrical Systems Korea06/2021 – 12/2024 (3 yrs 7 mos)

Site SQA Engineer · Gyeongju · Full-time · Left by internal transfer (ADAS division spin-off)

Field claim RCA — MDPS steering (sensor product)

- Steering field issue registered as a customer major quality agenda. With no R&D support on site, self-studied the Hall IC data sheet and EEPROM structure, analyzed returned parts (PGSIS-2 tool, oscilloscope reproduction tests).
- Found a Threshold Margin Level written in an unstable band causing parity errors; proved the customer's re-writing voltage drop as root cause in a joint investigation with the customer and IC maker — **~KRW 26M** claim cost defended.
- Set up EEPROM data-integrity sampling inspection, limit samples, and inspection standards.

New supplier APQP — SMT/PCBA resourcing (2022.12 – 2023.03)

- Led process audits, PV test selection, FDPR, Cp/Cpk and MSA, and FMEA/Control Plan updates for dual-sourcing a PCBA supplier.
- At line try-out, traced 100% FCT failure to a wrong EEPROM write value (LP filter '100' instead of '011') via data-sheet analysis; added write-then-read 100% verification to the Control Plan — **45%** PPM reduction after SOP.

Production & system quality

- Led IATF 16949 and Valeo V5000 (TQ) audits in 2023 and 2024; certification maintained.
- New-mold ISIR approval for 3 injection parts (2024): PFMEA update, capability/MSA review, gauge specs — FDPR process defect rate **–10%**.
- TAS line top scrap (EOLT hysteresis): identified PCBA flux tilting and housing structure causes — **21%** PPM reduction.
- Handled supplier quality issues with global suppliers (Thailand, Slovakia, Germany) in English; recovered quality costs incl. ~KRW 60M and ~KRW 80M claims.

Dokkaebi Co., Ltd.10/2020 – 03/2021 (5 mos)

Product Designer · Gumi · Full-time · Left due to company closure

- Designed an e-kickboard body and safety accessories (wind shield, LED safety system) for a national R&D project; built 3D-printed prototypes and coordinated welding/machining vendors.

DIGEN Co., Ltd.08/2019 – 12/2019

Process Engineering Intern (Electronics) · Daegu

- Proved test logic with an Arduino current-measurement module, then led outsourced development of a 100% automated inspection system with real-time Bluetooth data monitoring.

SKILLS

Quality methods

APQP, PPAP/ISIR, PFMEA, Control Plan, FDPR, SPC (Cp/Cpk), MSA/Gage R&R, FTA, PDCA, 8D, 4M change

Standards & audits

IATF 16949, Valeo V5000 / 5 Star, supplier process audit

Electronics

PCBA/SMT, FCT/EOLT, Hall IC EEPROM analysis (PGSIS-2), oscilloscope

Systems

SAP, Apriso (MES)

Data & automation

Excel VBA, Python (pandas), Google Apps Script, HTML/JS dashboards, Supabase/Vercel, Power Automate, Arduino

EDUCATION

Kumoh National Institute of Technology — B.S. Mechanical Engineering (GPA 3.81/4.5)03/2014 – 02/2020

ADDITIONAL

- ROK Navy (2015 – 2016), Cheonghae Unit anti-piracy escort mission off Somalia · OPIc English IM2 (Aug 2025)